

TEZPUR UNIVERSITY
Test V, Autumn 2015
C0 504 Natural Language Processing

Marks : 25

Time : 30 minutes

The figures in the right-hand margin indicate marks for the individual question.

1. Give examples of homonymy, homophones, homographs, polysemy and synonymy. 5
2. Describe a situation in linguistic expressions analysis where knowledge of *hyponymy* is required. 2
3. Consider the two sentences- (i) Kamal broke a glass, and (ii) The wind broke a glass. How are the roles of *Kamal* and *the wind* different in the two sentences? 3
4. What does *inverse document frequency* indicate? How is it used in information retrieval? 2 + 2

5. How can *selectional restriction* be used in word sense disambiguation? Give an example. What are some difficulties in using selectional restriction for WSD? 2 + 3

6. What can be the role of a dictionary in word sense disambiguation? 2

7. The word *bank* may have different senses, and the words *forest* and *vegetation* can be used to mean the same in some contexts. How would this effect *precision* and *recall* in information retrieval? 2

8. What is the reference resolution issue in the following sentences- 2
Anil drove the car slowly over the bridge. It was recently repaired.